MAWHIBA METAMINDS · LEVEL 1

# DVLD

## Driving & Vehicle License Management System

PROJECT PRESENTATION | APPENDIX D

C#

WINFORMS

ADO.NET

SQL SERVER

Mawhiba MetaMinds (M3) • Imam Mohammad Ibn Saud Islamic University

# The problem

Licensing procedures become difficult to manage when records and rules are fragmented.

## Fragmented records

Paper-driven or disconnected records can make identity, applications and licensing history difficult to manage.

## Manual verification

Eligibility, prerequisite tests and application status require consistent checks to avoid mistakes.

## Operational complexity

Issuance, renewals, replacements and detentions involve connected workflows and data.

# Our solution

One desktop system for administrative control of the driving-license lifecycle.

## DVLD unifies core licensing operations

- Centralized people, drivers and user records
- License applications and multi-stage testing
- Issuance, renewal, replacement and detention tracking
- Business rules and status checks in a dedicated layer

## Project objective

A structured administrative prototype that demonstrates maintainable software architecture, relational data modeling and more consistent workflows.

# What the system manages

The main modules documented in the final project report.

01 People & users

02 Drivers & records

03 Local applications

04 Vision / Written / Street tests

05 License issuance & renewal

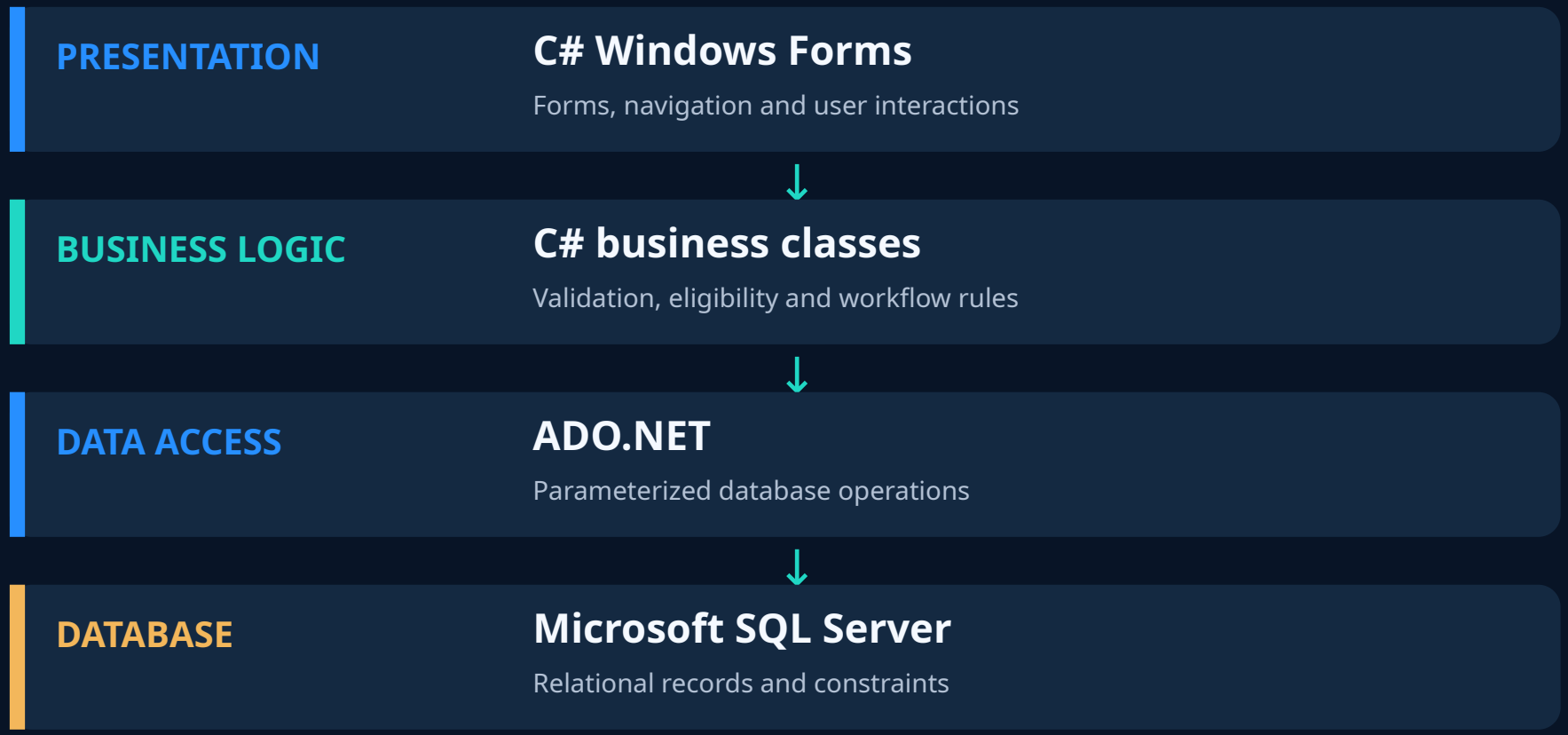
06 Replacements & detentions

07 International licensing

08 Fees & application status

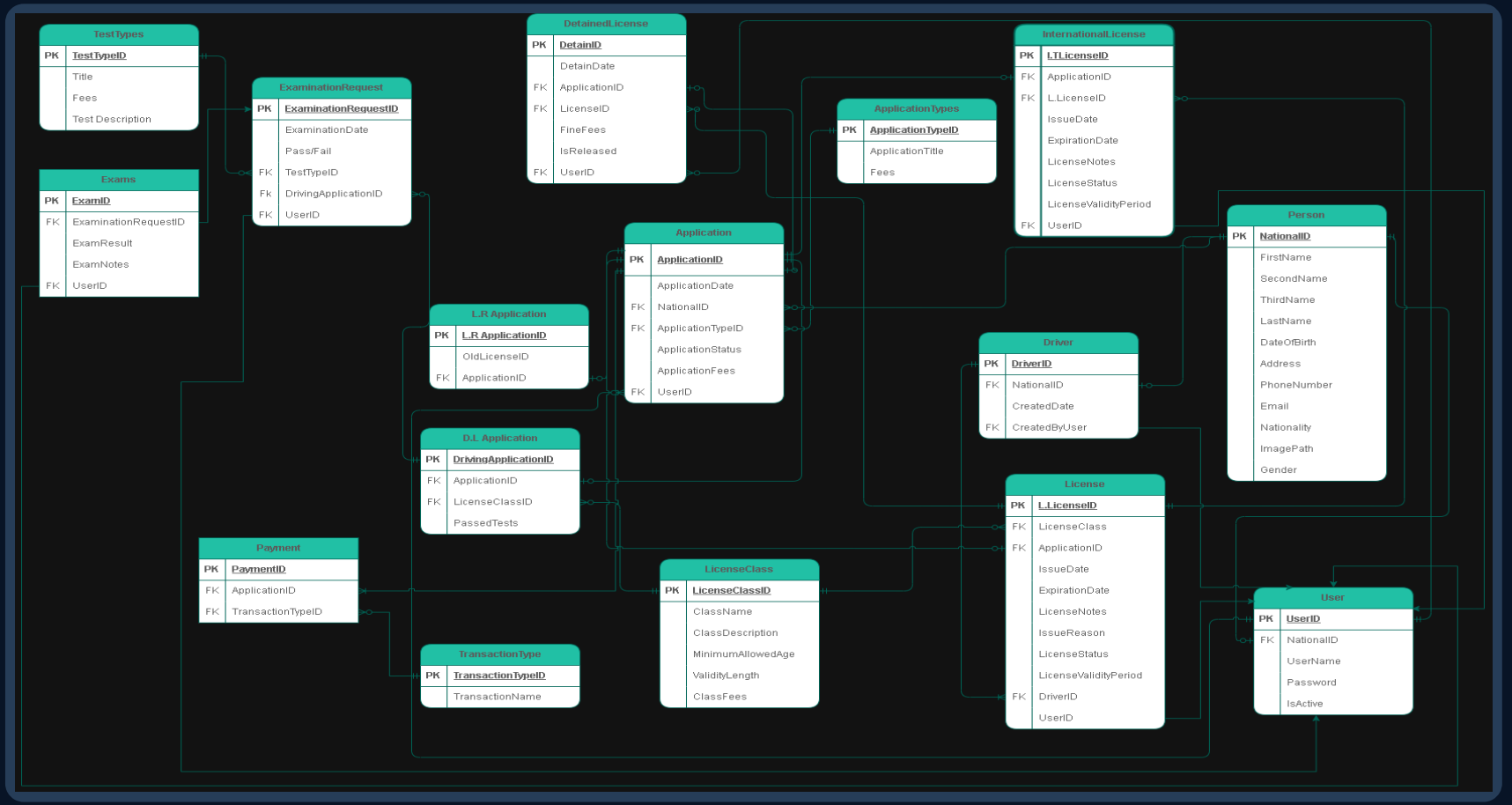
# Architecture: clean separation

A three-tier application design, connected to a relational SQL Server database.



# Database design

A relational schema connects applicants, tests, applications, drivers and licenses.




## ERD overview

Primary and foreign keys link the main entities and support consistent processing across related licensing records.

# A licensing journey

A structured sequence from application to license issuance.

**01**
**Applicant record**

**02**
**Local application**

**03**
**Vision test**

**04**
**Written test**

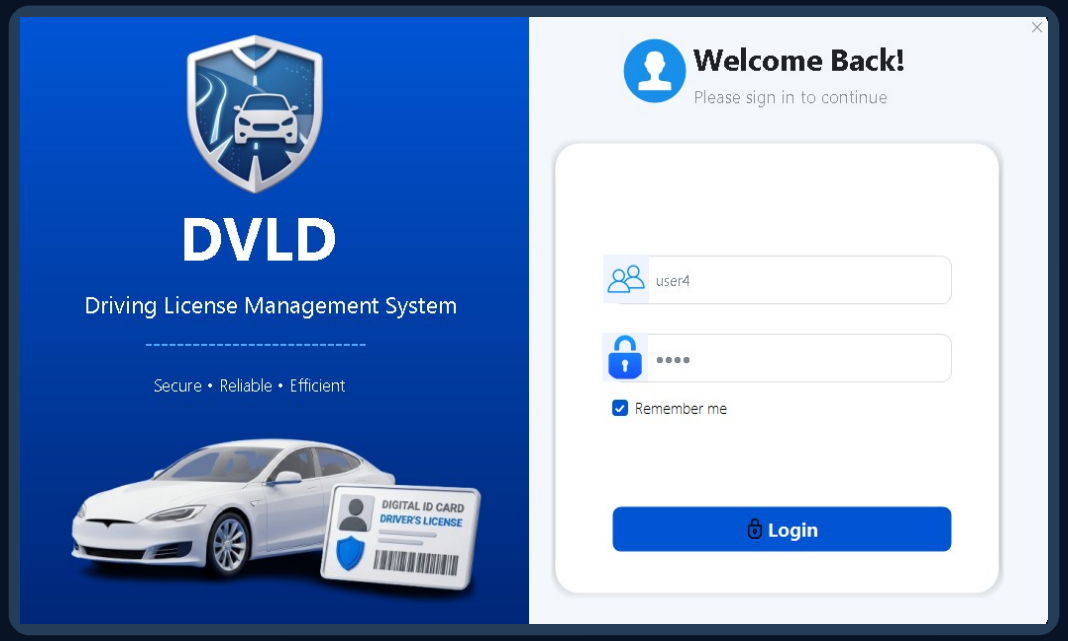
**05**
**Street test**

**06**
**Issue license**

Progression is governed by the application status and prerequisite checks implemented in business logic.

# The desktop experience

Login and home navigation.




Authentication interface



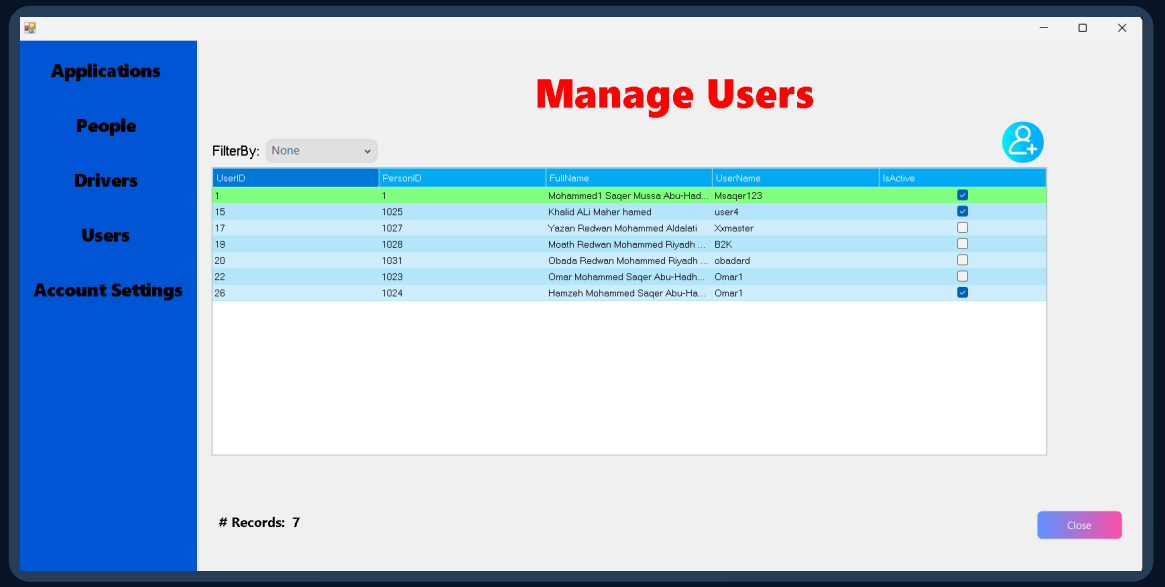

Main application navigation

# Administrative operations

Real application screens demonstrate record and user management.




People management

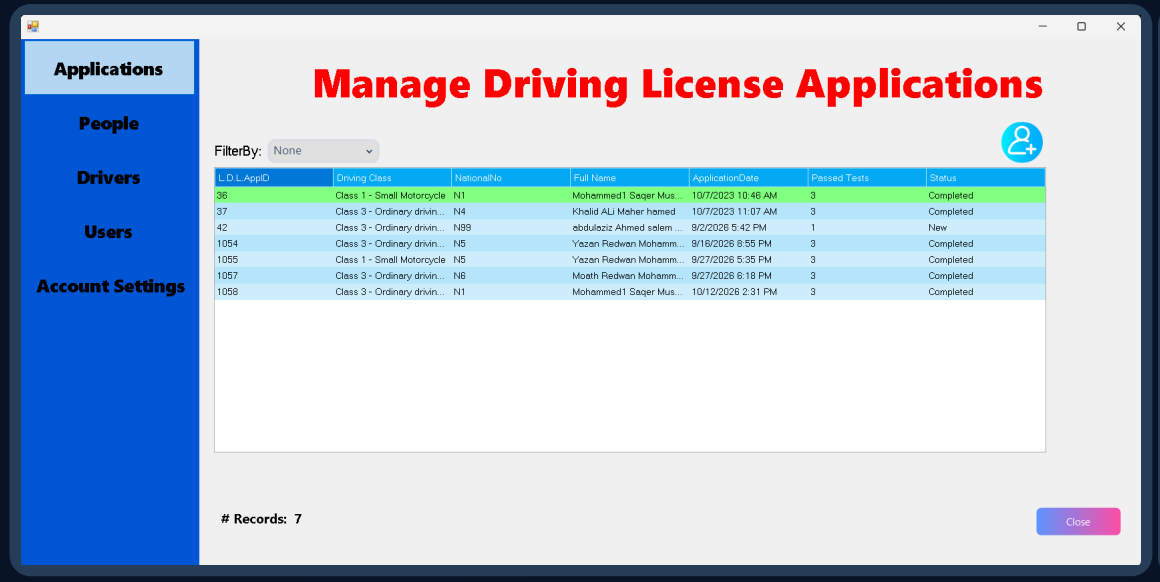


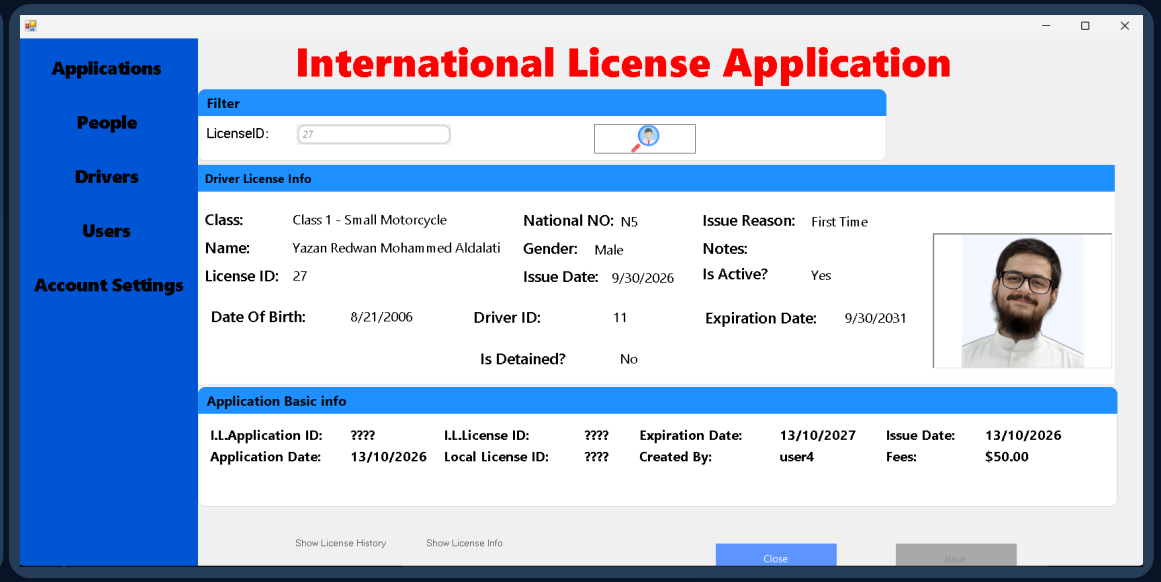

User management

Additional driver and application screens are documented in Appendix B.

# Licensing workflows

Application management and international license processing.




## Driving-license applications

## International license application

# How the project was developed

Development phases described in the M3 report.

| | | |
|---|---|---|
| 01 | **Database modeling** | ERD and SQL schema |
| 02 | **UI/UX prototyping** | Figma designs and navigation |
| 03 | **Core implementation** | Presentation, BLL and DAL |
| 04 | **Workflow integration** | Applications, tests and licensing |
| 05 | **Review and testing** | Mentor review and functional checks |

# Project team & documented roles

Responsibilities are presented as listed in the submitted project report.

## Yazan Redwan Aldalati

Team Leader
Full-stack development
System architecture & database lead

## Mohammed Ammar Altryfee

UI/UX design
Figma prototyping
Frontend interface planning

## Moath Redwan Aldalati

Database design
System design
Feature mapping

**Faculty mentor: Dr. Mostafa Ibrahim**

# What the project demonstrates

A working software engineering case study, with room for future extension.

## Deliverable

A C# WinForms application with major administration and licensing modules connected to SQL Server.

## Engineering value

Separation of concerns, relational modeling, business-rule enforcement and repeatable workflows.

## Potential impact

A prototype reference for digitizing and organizing administrative workflows, not a deployed public service.

# Challenges & lessons

Technical decisions and project-management insights from the submitted report.

- Sequential test prerequisites require reliable state validation.
- Separation of UI, business logic and data access makes changes easier to isolate.
- Database relationships are central to application consistency.
- A milestone-driven scope helps prioritize the core licensing workflow.
- Design handoffs need clear documentation and technical review.

# DVLD

**From administrative complexity to structured digital workflows.**

## Thank you

Source code: github.com/CYZNS/DVLD-Management-System